

CS 111

Foundations of Program Design
Week 4, Wednesday Lecture

Quiz 3

Loop rewriting

Recall the FizzBuzz program:

```
int i = 0;
while (i < 15) {
    // I am a comment, arbitrary code goes here ...
    i += 1;
}
```

Using the `for` syntax, we may rewrite this `while` loop to avoid splitting the definition of the iterator variable `i` over three lines.

```
for (int i = 0; i < 15; i += 1) {
}
```

Further exercises are available from the mooc.fi text, part 2-3:

<https://java-programming.mooc.fi/part-2/3-more-loops>

Topics

Dividing the program

Random sampling / Argument passing

Arrays

Array initialization / Byte length

Random sampling

$x*x + y*y \leq 1$ is the formula for a unit circle centered at (0,0). Using `Math.random()` to sample points (x,y) at random, we can measure the area of the circle relative to the unit square.

```
public class MonteCarloCircleArea {
    public static void main(String[] args) {
        int totalPoints = 100;
        int pointsInsideCircle = 0;

        for (int i = 0; i < totalPoints; i++) {
            // Generate random pairs x and y between 0 and 1.
            double x = Math.random();
            double y = Math.random();

            // Is this point inside the quarter-circle?
            if (x * x + y * y <= 1) {
                pointsInsideCircle++;
            }
        }
        System.out.println(pointsInsideCircle); // total hits in the quarter-circle
    }
}
```

Argument passing

In order to perform statistical sampling with different numbers of points, we can parameterize the `sample` method. Instead of declaring `totalPoints` in the function body, we'll turn it into an *argument* of the `sample` method.

```
public class MonteCarloCircleArea {
    public static void sample(int totalPoints) {
        long pointsInsideCircle = 0;

        for (long i = 0; i < totalPoints; i++) {
            // Generate random pairs x and y between 0 and 1.
            double x = Math.random();
            double y = Math.random();

            // Is this point inside the quarter-circle?
            if (x * x + y * y <= 1) {
                pointsInsideCircle++;
            }
        }
        return pointsInsideCircle;
    }

    public static void main(String[] args) {
        System.out.println(sample(100));
        System.out.println(sample(10000)); // adding samples increases accuracy
    }
}
```

Array initialization

Java arrays are useful to store sequences of values, such as the various inputs and outputs of `for` and `while` loops.

This String array is initialized to a literal value, written into the program. Next week, we'll learn how to read arrays from files saved next to your Java source code in the working directory.

```
public class DaysOfWeek {  
    public static void main(String[] args) {  
        String[] daysOfWeek = {"Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"};  
        String message = daysOfWeek[0];  
        for (int i = 1; i < 7; i += 1) {  
            message += " " + daysOfWeek[i];  
        }  
        System.out.println(message);  
    }  
}
```

Start reading mooc.fi 3-3:

<https://java-programming.mooc.fi/part-3/3-arrays>

Byte length

For some problems, you may need to declare an array without initializing it. But Java needs to allocate enough memory for the array, so you need to provide its length as well as its name.

```
public class SamplePoints {  
    public static void main(String[] args) {  
        double[] samplePoints = new double[100]; // note 'new' keyword  
        for (int i = 0; i < 100; i += 1) {  
            samplePoints[i] = Math.random() * 100;  
        }  
        System.out.println(samplePoints);  
    }  
}
```

Optionally, you can read the Java type specification describing the width of each primitive type in bytes of memory:

<https://docs.oracle.com/javase/tutorial/java/nutsandbolts/datatypes.html>